

Central Bank Digital Currency Spillovers^{*}

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Abstract

This paper analyzes the cross-border spillover effects of central bank digital currency (CBDC). In a two-country model of the financial accelerator, I show that the introduction of CBDC materially changes the international spillover effects of country-specific monetary and real shocks, confirming the findings of earlier studies. In addition, I show that CBDC can better insulate banks across countries in response to negative shocks, thereby acting as an international (negative) shock absorber. In this regard, CBDC can prove to be particularly beneficial for bank stability, especially during economic downturns.

JEL Codes: E44, E52, E58, F41

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1 INTRODUCTION

Following the [Bank of Japan's](#) and [ECB's](#) press releases on the pilot phase of a central bank digital currency (CBDC), several prominent institutions (Bank of England, Federal Reserve Board) have initiated efforts to evaluate the merits of a digital transformation in the conduct of monetary policy. While most studies have extensively examined the closed-economy implications of a digital currency adoption, significantly less attention has been given to its open-economy effects.

This paper aims to analyze the domestic and international transmission of country-specific shocks in the presence of a central bank digital currency. To this end, I integrate a central bank digital currency in a two-country, open-economy New Keynesian model with nominal and real rigidities, and financial frictions to explore how domestic real and nominal shocks propagate across borders. My findings are twofold. First, introducing CBDC amplifies the international transmission of domestic shocks, corroborating the key predictions of [Minesso et al. \(2022\)](#). Second, and novel relative to the existing literature, the introduction of CBDC is shown to better insulate domestic and foreign banks from adverse shocks to total factor productivity. In particular, in response to a negative domestic productivity shock, the model with CBDC generates milder contractions in bank profitability and net worth, while leverage and lending spreads raise by less. As such, CBDC can act as an international (negative) shock absorber.

Relative to [Minesso et al. \(2022\)](#), a key methodological contribution of this paper is the development of a unified framework with a realistic competitive retail banking sector that allows one to jointly analyze (i) the extent to which adding a digital monetary instrument reshapes the transmission mechanism of conventional monetary policy, (ii) the effects and the cross-country propagation of shocks on bank characteristics (profitability, net worth, leverage, credit spreads), and (iii) the international transmission of unconventional central bank balance sheet interventions.¹ In the model, the central bank finances the acquisition of government and private debt through the issuance of reserves, cash, and CBDC. The demand for cash and CBDC comes from households' need for liquid payment instruments to finance consumption spending. Retail banks intermediate funds between households, productive firms, and the government. To extend loans to non-financial firms, financial intermediaries use own equity and external financing through deposit

¹The latter, while possible within the context of the present model, is not pursued in this paper.

contracts. To limit the ability of a bank to raise all the necessary liquidity to finance its activities, a principal agent friction in the spirit of [Gertler and Kiyotaki \(2010\)](#) is introduced. In particular, every period the bank can divert funds for personal gain. This limits the amount of deposits households are willing to hold within the banking industry, effectively imposing an endogenous capital constraint on banks. This constraint generates endogenous amplification of shocks, and can give real effects to central bank balance sheet policies. Therefore, the model provides a rich setting to quantify and analyze both types of conventional and unconventional monetary policies.

Literature This paper focuses on the rapidly evolving body of research exploring the positive and normative implications of central bank digital currencies (CBDCs). In recent years, numerous researchers — primarily from major policy institutions, with some contributions from academia — have undertaken early assessments of the benefits and potential risks associated with this policy initiative. Broadly, the existing literature can be categorized into three key dimensions through which CBDCs may impact the macroeconomy, offering distinct perspectives on their potential influence.

The first line of research stresses the effects of CBDC on bank funding (dis)intermediation. A concern central in policy discussions is the possibility of households substituting sizeable amounts of bank deposits for newly issued digital currencies, causing a disintermediation in the banking sector. [Whited et al. \(2022\)](#) employ an equilibrium model of banking industry dynamics with imperfectly competitive private banks to estimate that the introduction of unremunerated CBDC disintermediates banks so long as bank deposits compete with CBDC. [Williamson \(2022a\)](#) demonstrates that the issuance of CBDC induces a welfare-improving disintermediation as it relocates safe assets from competitive commercial banks — that being subject to principal agent frictions may end up with inefficient levels of collateralizable assets to back liabilities in equilibrium — to the central bank. [Keister and Sanches \(2023\)](#) develop a New-Monetarist model with perfect banking competition and financial frictions, in which positive CBDC holdings are guaranteed in equilibrium only to the extent that deposits incur a liquidity premium that disintermediates banks. In the model, even though CBDC crowds out bank deposit funding, it is socially desirable from a welfare perspective as it improves the efficiency in exchange. [Chiu et al. \(2023\)](#) use a modified version of the [Lagos and Wright \(2005\)](#) model with

bank market power to highlight that the effect of the CBDC rate on bank intermediation is non-monotonic. A moderate remuneration crowds in private deposits as it forces banks to offer more attractive deposit contracts, whereas a generous remuneration causes an overall drop in loan provision as banks can not match the deposit rate with that paid on CBDC, raising lending rates to mitigate losses.

The second line of research examines the implications of CBDC for financial (bank) stability. Early papers in this direction are [Williamson \(2022b\)](#) and [Whited et al. \(2022\)](#). [Williamson \(2022b\)](#) evaluates flight-to-safety considerations and corroborates the view that CBDC increases the risk of financial panics (but makes depositors better-off during a panic). This view is shared by [Whited et al. \(2022\)](#) who argue that CBDC increases banks' exposure to interest rate risk for it curtails the deposit base of banks financing long-term projects. In a similar vein, [Schilling et al. \(2024\)](#) and [Bidder et al. \(2024\)](#) document that in times of banking distress, CBDC can facilitate runs. On the more conservative side, [Ahnert et al. \(2023\)](#) employs a [Diamond and Dybvig \(1983\)](#) model to emphasize the U-shaped relation between financial panics and CBDC remuneration. I differentiate from the aforementioned papers in that I provide a complete characterization of the transitional dynamics in response to various shocks, instead of focusing exclusively on partial equilibrium or steady state analyses.²

My paper principally relates to a third strand of the literature that explores the macroeconomic effects of CBDCs using quantitative DSGE models. Previous studies have developed versions of the canonical New Keynesian model augmented with banks and various nominal and real rigidities, in which monetary policy operates solely through conventional interest rate (and Taylor-type) rules. [Barrdear and Kumhof \(2022\)](#) demonstrates that incorporating a CBDC backed by government debt into a New Keynesian model can lead to permanent output gains while enhancing banks' role in financial intermediation. [Burton et al. \(2024\)](#) calibrate a DSGE model for the euro area to identify the sources driving CBDC's impact on the broader economy. While a CBDC expands the central bank balance sheet and positively influences private consumption, the partial substitution of bank deposits with central bank funding contracts credit supply and suppresses steady-state output. [Abad et al. \(2025\)](#) setup a DSGE model with heterogeneous banks to investigate potential disruptions in the functionality of the interbank market in the presence of

²An exception is [Bidder et al. \(2024\)](#) who also study the effects of CBDC in the transition phase.

CBDC. They show that even though a digital currency may exhaust deposit funding for banks, the endogenous response of the central bank’s operational framework is sufficient to safeguard a steady flow of credit to the real sector. Contrary to the previous papers, I put forward a model of direct central bank intermediation ([Gertler and Karadi, 2013](#)), in which households solve a portfolio choice problem over assets with unique intrinsic features and an explicitly microfounded demand for liquid payment instruments, that allows to trace out the significance of CBDC when monetary policy operates in *conventional* and *unconventional times* (by means of large scale asset purchases). Closest to my paper is the work of [Minesso et al. \(2022\)](#), who utilize a two-country DSGE model with banks functioning in a stylized fashion, to address the global spillovers from a domestically issued — but internationally held — CBDC. Their analysis refines the uncovered interest parity condition by linking the foreign interest rate to a marked-up CBDC remuneration. This refinement heightens the sensitivity of the real exchange rate to exogenous shocks, amplifying spillover effects. Building on and extending [Minesso et al. \(2022\)](#), my model introduces two key innovations. First, I incorporate a formal representation of the commercial banking sector, where financial intermediaries with limited commitment ([Gertler and Kiyotaki, 2010](#)) extend loans to private firms and the government, financed through net worth and short-term debt.³ Second, I develop a more realistic (and practical) portrayal of monetary policy operations, wherein the central bank actively adjusts its balance sheet by issuing banknotes, reserves, and central bank digital currency. This integrated framework can be used to assess the propagation of idiosyncratic real and nominal shocks within and across borders, and the implications for financial stability; it can also potentially serve as a benchmark to highlight the real economic effects of central bank balance sheet adjustments, and facilitate an analysis of the cross-border implications of domestic asset purchase policies in response to country-specific shocks. In this regard, I also touch upon the literature on the open economy implications of national unconventional policies ([Dedola et al., 2013](#); [Bluwstein and Canova, 2018](#); [Hohberger et al., 2019](#); [Cova et al., 2019](#); [Alpanda and Kabaca, 2020](#); [Kolasa and Wesołowski, 2020](#); [Dedola et al., 2021](#)).

³Short term debt issued by private financial intermediaries takes the form of one-period (time) deposit contracts.

2 OPEN ECONOMY MODEL

The model builds on the seminal 2-country frameworks of [Backus et al. \(1992\)](#) and [Eichenbaum et al. \(2021\)](#) augmented with a rich banking sector subject to financial frictions ([Dedola et al., 2013](#)). To this, I add a new monetary instrument, a central bank digital currency ([Minesso et al., 2022](#)).

The world economy consists of two symmetric production economies (home and foreign) that exchange in good and financial asset markets. Different from [Backus et al. \(1992\)](#), frictions in asset trade between countries make international asset markets incomplete and violate the uncovered interest parity (UIP) condition. Full risk-sharing is preserved within countries.

There are five agents in each economy: *(i)* households, *(ii)* firms, *(iii)* financial intermediaries (henceforth banks), *(iv)* a monetary authority (central bank), *(v)* and a fiscal authority (Treasury).

Households consume the final good, save in domestic and foreign assets, and supply labor. Portfolio allocation spans five types of financial instruments with unique characteristics: domestic short term deposits remunerated at the risk free rate, but which do not provide liquidity services;⁴ domestic and foreign short and long term government bonds remunerated at the respective risk free and long term rates, which provide no liquidity services either; and cash in hand, which is unremunerated but provides liquidity services. To account for risks associated with holding large amounts of physical cash, I assume that cash holdings incur (nominal) storage costs.

Banks transfer funds between households, productive firms, and the government. As is standard in the literature that assesses the power of unconventional policies ([Gertler and Kiyotaki, 2010](#); [Gertler and Karadi, 2013](#)), I allow for an agency problem that places a limit on the amount of funds a bank can raise from households, and introduces an (endogenous) incentive constraint. A binding incentive constraint breaks [Wallace \(1981\)](#) neutrality and enables central bank interventions to have real effects. In the model, the agency friction interacts with a variety of real and nominal frictions. On the demand side, households exhibit (external) habits in consumption, and face portfolio adjustment costs when holding domestic short and long term bonds beyond a frictionless benchmark. Portfolio costs also apply to cross-border holdings of short term government debt.

⁴One can think of them as time deposits that only become available at maturity.

On the supply side, final goods producers interact with capital and intermediate firms to produce final output, which is sold within and across borders. Capital firms are subject to adjustment costs when varying (gross) investment between time periods (Christiano et al., 2005). Intermediate firms face Calvo (1983) type price rigidities when invoicing domestically consumed and exported goods. Nominal and real frictions are necessary to fit the model to business cycle statistics — such as the autocorrelation between aggregate output, consumption, and investment — and generate the empirically plausible hump-shaped impulse responses. Similarly, the principal agent friction in the banking sector is key to replicate the main macroeconomic effects of central bank balance sheet interventions.

The model closes with two central authorities. A central bank performs conventional interest rate policy in normal times and expands its toolkit by directly intervening in private and public asset markets during periods of financial distress. The Treasury issues short and long term bonds to finance public spending, and adjusts taxes to ensure fiscal solvency.

Central in my setup is the assumption that — in addition to cash and reserves — the central bank in the home country can issue a central bank digital currency. Digital currency is a form of electronic money, and is not subject to holding costs.⁵ It possesses several attractive features; it is perfectly liquid and can be used as a means of instant payments, and depending on its design it can — potentially — be remunerated. CBDC is available to home and foreign households, which provides another source of financial integration. Moreover, similar to cash and reserves, it constitutes a liability for the central bank that is used to finance monetary policy operations. Together with cash, they represent central bank (public) money and, by facilitating transactions, enter household utility in a *money-in-utility* fashion.

In the remainder of the section I present the problem of the main agents in the *domestic* country.⁶ The problem is symmetric for the foreign economy except for the issuance of central bank digital currency. Foreign variables are marked by an asterisk.

⁵Assuming away storage costs for electronic money makes CBDC relatively cheaper for households as a liquid payment instrument. As such, CBDC becomes more attractive than cash, all else equal. This — together with the assumption that CBDC can potentially be remunerated — implies that when a shock hits, portfolio rebalancing by households will induce a stronger substitution away from cash toward CBDC. As shown in Section 3, this is the main reason behind the exchange rate overshooting and the international amplification of country-specific shocks.

⁶I will use the words *domestic* and *home* interchangeably.

2.1 Set up

Long term bonds Long term bonds are modeled as in [Hatchondo and Martinez \(2009\)](#). A government bond B_t^L issued in period t at (real) price Q_t is a security that obligates the issuer to a sequence of geometrically decaying (real) coupon payments ψ , $(1 - \varrho)\psi$, $(1 - \varrho)^2\psi \dots$ where $\varrho \in [0, 1]$ is the decay parameter for cash flows and captures the maturity structure of government debt; in the two polar cases, $\varrho = 1$ corresponds to short term debt (one period maturity), and $\varrho = 0$ represents a perpetuity (consol). The attractive feature of long term bonds is that current bond issuances are sufficient to pin down the stream of future debt obligations stemming from past issuances. Formally, I can express this as

$$B_{t-j}^L = \sum_{j=1}^{\infty} (1 - \varrho)^j B_t^L.$$

A bond issued in period $t - j$ is equivalent to $(1 - \varrho)^j$ bonds issued in period t at (real) price Q_t , that pay a sequence of known future cash flows. In that sense, B_{t-j}^L represents long term debt in equivalent newly issued long term bonds B_t^L . All future payment obligations are summarized in a single state variable, B_{t-j}^L .

Households The domestic economy is populated by a continuum of infinitely lived identical households in the unit interval. As in [Minesso et al. \(2022\)](#) the representative household has preferences given by the instantaneous utility function

$$U_t = e_t^c \log(C_t - hC_{t-1}) - \chi \frac{L_t^{1+\varphi}}{1+\varphi} - \chi_{DC} \mathcal{G} \left(\frac{M_t}{M_t + DC_t^H}, \Gamma \right)^2, \quad (1)$$

where C stands for consumption, L hours worked, M cash holdings, and DC^H holdings of central bank digital currency⁷; h captures external consumption habits; φ is the inverse elasticity of labor supply to variations in the wage rate; $\chi > 0$ and $\chi_{DC} > 0$ are scaling parameters; e^c denotes consumption preferences and follows an AR(1) process. Instantaneous utility U is twice differentiable, strictly increasing and concave in consumption, strictly decreasing and convex in labor supply, and decreasing and convex in the wedge between the actual allocation of payment instruments and the preferable mix, defined by

⁷Throughout the paper superscript H refers to variables originating from home country, F to variables originating from the foreign economy, and h to variables held within the household sector.

$\mathcal{G} = \left(\Gamma - \frac{M_t}{M_t + DC_t^H} \right)^2$. Function $\mathcal{G}$ has a global minimum at $\Gamma \equiv \frac{M_t}{M_t + DC_t^H}$, and $\Gamma \in [0, 1]$ reflects household preferences over payment instruments. This functional form implies that when the realized relative allocation of cash $\frac{M_t}{M_t + DC_t^H}$ deviates from the desirable payment instrument mix Γ , the household incurs a cost quadratic in the deviation.

Optimizing households are forward-looking and smooth consumption by trading in markets for government securities. The portfolio allocation of the representative household consists of short term bank deposits D_t , (real) balances of domestic and foreign-currency-denominated short term government bonds B_t^H and B_t^F , (real) balances of domestic-currency-denominated long term government debt $Q_t B_t^{L,h}$ ($B_t^{L,h}$ are units of long term bonds each priced at the real price Q_t), physical cash M_t (expressed in real terms), and central bank digital currency DC_t^H (expressed in real terms). Following [Gertler and Karadi \(2013\)](#) I assume that households are less efficient than banks in managing investment projects of long maturity when the scale of investment is such that it requires broker expertise. This inefficiency is captured by a convex transaction cost function $\frac{1}{2}\kappa \left(B_t^{L,H,h} - \bar{B}^{L,H,h} \right)^2$ that equally penalizes households for holding bonds above or below a frictionless level, where $\bar{B}^{L,H,h}$ is the capacity households can effectively manage without any expertise. Portfolio costs of this kind have been used extensively in models with unconventional central bank intermediation as they break the substitutability between assets, leading to positive excess returns in equilibrium ([Gertler and Karadi, 2013](#); [Hohberger et al., 2019](#); [Guo et al., 2023](#)).

The flow budget constraint of the representative household is given by

$$\begin{aligned} C_t + D_t + M_t + B_t^H + Q_t \left[B_t^{L,H,h} + \frac{1}{2}\kappa \left(B_t^{L,H,h} - \bar{B}^{L,H,h} \right)^2 \right] + \left(NER_t \frac{P_t^*}{P_t} \right) B_t^F \\ + T_t + DC_t^H \leq W_t L_t + R_t^D D_{t-1} + R_t B_{t-1}^H + R_t^L Q_{t-1} B_{t-1}^{L,H,h} + \left(NER_t \frac{P_t^*}{P_t} \right) R_t^* B_{t-1}^F \quad (2) \\ + R_t^{DC} DC_{t-1}^H - \frac{\phi^B}{2} \left(NER_t \frac{P_t^*}{P_t} B_t^F \right)^2 + \frac{\xi^\$}{\pi_t} M_{t-1} + Div_t. \end{aligned}$$

P_t and P_t^* are domestic and foreign price levels. Each period households consume (C_t), make deposits to private banks (D_t), hold liquid means of payments in the form of physical cash in hand (M_t) and digital currency (DC_t^H), and pay real lump-sum taxes (T_t); they also invest in domestic and foreign short term government bonds (B_t^H and B_t^F , where the nominal exchange rate NER_t is defined as the price of foreign currency), and

in domestic long term government bonds ($B_t^{L,H,h}$). Expenditures are financed through labor and dividend income ($W_t L_t$ and Div_t),⁸ interest income on deposits ($R_t^D D_{t-1}$); domestic and foreign short term bonds ($R_t B_{t-1}^H$ and $R_t^* (NER_t \frac{P_t^*}{P_t}) B_{t-1}^F$); domestic long term bonds ($R_t^L Q_{t-1} B_{t-1}^{L,H,h}$); and domestic CBDC ($R_t^{DC} DC_{t-1}^H$), net the the storage cost of cash and the portfolio cost of holding foreign short term bonds. W_t is the real wage, and R_t , R_t^* , R_t^D and R_t^L the gross real returns on domestic and foreign short term bonds, deposits, and domestic long term debt. Nominal storage cost $\xi^\$ \in [0, 1]$ is assumed to scale linearly with the amount of cash. I also assume frictions in international financial markets $\left(\frac{\phi^B}{2} \left(NER_t \frac{P_t^*}{P_t} B_t^F \right)^2 \right)$ to induce deviations from the uncovered interest parity, as empirically supported by [Backus and Smith \(1993\)](#) and [Bansal \(1997\)](#).⁹

To purchase the aggregate final good, households need liquidity. Liquidity is a constant elasticity of substitution (CES) aggregate of cash and CBDC, defined as

$$\mathcal{F}_t = \left[\mu^\$ M_t^{1-\rho_{DC}} + \mu^{DC} (DC_t^H)^{1-\rho_{DC}} \right]^{\frac{1}{1-\rho_{DC}}}, \quad (3)$$

where $\mu^\$$ and μ^{DC} are the weights of each payment instrument in the CES aggregator and capture household preferences over liquidity services; ρ_{DC} is the inverse of the elasticity of substitution between cash and digital money. The CES specification allows to treat cash and CBDC as imperfect substitutes. This is because CBDC possesses several attractive features; similar to cash — but unlike bonds and deposits — it can be used as a means of payment for the purchase of final goods; and unlike cash — but like bonds and deposits — it can earn a positive return (if remunerated). In addition to its synthetic nature, households may choose to hold CBDC owing to its intrinsic characteristics. For instance, even without earning interest, some households may prefer to hold digital currency because they can make in-store payments conveniently using their smartphones (touchless payments) or conduct online transactions seamlessly. Moreover, it can be used for cross-border transactions, or even for domestic transactions invoiced in a foreign currency.

⁸Dividend income (Div_t) is the sum of distributed earnings accrued to households from the ownership of financial and non financial firms (i.e. intermediate and capital firms), after accounting for the transfers to new bankers.

⁹Convex adjustment costs on holdings of foreign assets is a necessary condition for the stationarity of the net foreign asset position (NFA), and, subsequently, of the real exchange rate ([Schmitt-Grohé and Uribe, 2003](#)).

It follows that the cash-in-advance constraint can be expressed as

$$C_t \leq \mathcal{F}_t. \quad (4)$$

The constraint always binds in equilibrium. As such, equation (4) limits consumption levels to the extent of available liquidity.¹⁰

The representative household chooses the sequence $\{C_t, L_t, D_t, B_t^H, B_t^F, B_t^{L,h}, M_t, DC_t^H\}_{t=0}^\infty$ to maximize lifetime utility subject to the budget constraint and the cash-in-advance constraint. The first order conditions are

$$\lambda_t + \gamma_t = \frac{e_t^c}{C_t - hC_{t-1}} - h\beta E_t \left(\frac{e_{t+1}^c}{C_{t+1} - hC_t} \right) \quad (5)$$

$$\lambda_t W_t = \chi L_t^\varphi \quad (6)$$

$$1 = E_t \left(\beta \frac{\lambda_{t+1}}{\lambda_t} R_{t+1}^D \right) \quad (7)$$

$$1 = E_t \left(\beta \frac{\lambda_{t+1}}{\lambda_t} R_{t+1}^L \right) \quad (8)$$

$$1 + \phi^B \left(NER_t \frac{P_t^*}{P_t} \right) B_t^F = E_t \left[\beta \frac{\lambda_{t+1}}{\lambda_t} \frac{NER_{t+1}}{NER_t} \frac{\pi_{t+1}^*}{\pi_{t+1}} R_{t+1}^* \right] \quad (9)$$

$$B_t^{L,H,h} = \bar{B}^{L,H,h} + E_t \frac{\left(\beta \frac{\lambda_{t+1}}{\lambda_t} R_{t+1}^L - 1 \right)}{\kappa} \quad (10)$$

$$\begin{aligned} \lambda_t - \frac{\xi^\$}{\pi_{t+1}} E_t (\beta \lambda_{t+1}) &= (\mathcal{F}_t)^{\rho_{DC}} \mu^\$ (M_t)^{-\rho_{DC}} \gamma_t - \chi_{DC} 2 \left(\Gamma - \frac{M_t}{M_t + DC_t^H} \right) \\ &\times \left[\frac{M_t}{(M_t + DC_t^H)^2} - (M_t + DC_t^H)^{-1} \right] \end{aligned} \quad (11)$$

$$\begin{aligned} \lambda_t - E_t (\beta \lambda_{t+1} R_{t+1}^{DC}) &= (\mathcal{F}_t)^{\rho_{DC}} \mu^{DC} (DC_t^H)^{-\rho_{DC}} \gamma_t - \chi_{DC} 2 M_t \\ &\times \left(\Gamma - \frac{M_t}{M_t + DC_t^H} \right) (M_t + DC_t^H)^{-2}, \end{aligned} \quad (12)$$

where β is the household discount factor and E_t the expectations operator conditional on time- t information; $\{\lambda_t\}_{t=0}^\infty$ and $\{\gamma_t\}_{t=0}^\infty$ are the respective sequences of Lagrange multipliers for the budget and cash-in-advance constraints. Domestic and foreign inflation rates are denoted by π_t and π_t^* , respectively. The logarithmic AR(1) process for consumption

¹⁰Binding cash-in-advance constraint implies that households strictly prefer to invest in interest-bearing assets, and hold just as much liquidity as needed to finance current consumption.

preferences takes the form

$$\log(e_t^c) = (1 - \rho_{ec})\log(e^c) + \rho_{ec}\log(e_{t-1}^c) + \sigma_{ec}\epsilon_{ec}, \quad (13)$$

where $\sigma_{ec}\epsilon_{ec} \sim Niid(0, \sigma_{ec}^2)$ is the innovation to consumption preferences, σ_{ec} the standard deviation of the innovation, and ρ_{ec} the persistence of the process.

Equation (5) is the first order condition for consumption, and (6) the first order condition for labor supply. Deposit demand is given by equation (7). Equations (8) and (9) are the optimal choices for domestic and foreign short term government bond holdings, and (10) the optimal choice for domestic long term government debt. Optimality conditions for cash and CBDC are given by (11) and (12), respectively.

Equation (10) highlights a relationship between the demand for long term bonds, excess returns, and transaction costs. Demand beyond the frictionless capacity is increasing in excess returns relative to the portfolio cost parameter κ ; when the latter goes to infinity, households demand exactly the frictionless level. When the transaction cost is zero, interest rates are equalized (i.e. excess returns vanish) since households can engage in frictionless arbitrage between long and short term bonds.

The problem of the representative household in the foreign economy is fully symmetric except for the choice of CBDC — which is also subject to international financial frictions. Foreign demand for CBDC issued by the central bank in the home country is

$$\begin{aligned} (\mathcal{F}_t^*)^{\rho_{DC}^*} \mu^{DC*} \left(\frac{DC_t^{F*}}{NER_t \frac{P_t^*}{P_t}} \right)^{-\rho_{DC}^*} \gamma_t^* - \chi_{DC}^* 2M_t^* \left(\Gamma^* - \frac{M_t^*}{M_t^* + \frac{DC_t^{F*}}{NER_t \frac{P_t^*}{P_t}}} \right) \left(M_t^* + \frac{DC_t^{F*}}{NER_t \frac{P_t^*}{P_t}} \right)^{-2} \\ - \phi^{DC*} \lambda_t^* \frac{DC_t^{F*}}{NER_t \frac{P_t^*}{P_t}} = \lambda_t^* - E_t \left(\beta^* \lambda_{t+1}^* R_{t+1}^{DC} \frac{NER_t}{NER_{t+1}} \frac{\pi_{t+1}}{\pi_{t+1}^*} \right). \end{aligned} \quad (14)$$

Equation (14) defines the demand for CBDC by foreign households as a function of the liquid payment facilities it delivers, captured by γ_t^* (i.e. by how much an extra unit of CBDC relaxes the cash-in-advance constraint), its remuneration R_{t+1}^{DC} , preferences over other payment instruments, spot and forward nominal exchange rates NER_t and NER_{t+1} , and inflation rates π_{t+1}^* and π_{t+1} .

Financial Intermediaries A main difference of my model and that of [Minesso et al. \(2022\)](#) is the presence of a retail banking sector characterized by agency frictions, in the spirit of [Gertler and Kiyotaki \(2010\)](#) and [Gertler and Karadi \(2013\)](#).

A continuum of imperfectly competitive banks spans the unit interval $f \in [0, 1]$. Bank- f issues non-contingent short term deposit contracts $D_{f,t}$ to households, makes loans to domestic non-financial firms in exchange for capital claims $S_{f,t}^{H,b}$ (priced at q_t),¹¹ invests in domestic long term government bonds $B_{f,t}^{L,H,b}$ (priced at Q_t), and swaps securities for central bank reserves $RE_{f,t}$, when the central bank intervenes in private and government asset markets.

Let $N_{f,t}$ be the net worth of bank- f at the end of period t . I can express the balance sheet of bank- f as

$$q_t S_{f,t}^{H,b} + Q_t B_{f,t}^{L,H,b} + RE_{f,t} = N_{f,t} + D_{f,t}. \quad (15)$$

Bank's net worth evolves over time as the difference between gross interest paid on assets and the cost of liabilities (i.e. accumulated retained earnings)

$$N_{f,t+1} = R_{t+1}^k q_t S_{f,t}^{H,b} + R_{t+1}^L Q_t B_{f,t}^{L,H,b} + R_{t+1}^{re} RE_{f,t} - R_{t+1}^D D_{f,t}, \quad (16)$$

where R_t^{re} is the gross real interest on central bank reserves (i.e. reserve rate).

Let Z_t denote the net proceeds from holding firm equity, and δ the depreciation rate of capital. The stochastic gross real return on private capital claims is given by

$$R_{t+1}^k = \frac{Z_{t+1} + (1 - \delta)q_{t+1}}{q_t}. \quad (17)$$

In addition to holding private assets, banks invest in the market for long term government securities. In its general form, the real gross return on a long term government bond (bond rate) can be expressed as $R_{t+1}^L = (\psi + (1 - \varrho)Q_{t+1}) / Q_t$. For simplicity, I assume that long term bonds are consols ($\varrho = 0$) that pay every period a constant coupon (ψ) normalized to a unit of the final good. It follows that the bond rate reduces to

$$R_{t+1}^L = \frac{1 + Q_{t+1}}{Q_t}. \quad (18)$$

Banks are owned by households who receive any final period profits. The planning

¹¹Contrary to [Minesso et al. \(2022\)](#), who assume that banks do not hold capital.

horizon of each bank is finite, and banks can exit the industry every period with probability $1 - \sigma_B$, $\sigma_B \in [0, 1)$. This is to prevent banks from retaining enough net worth to self-finance investment projects, thereby relaxing any financing constraints. Exiting bankers are replaced by an equal mass of new entrants endowed — by households — with startup funds .

Bank's problem. The problem of each individual bank f is to maximize the present value of expected future dividends

$$V_{f,t} = E_t \sum_{i=0}^{\infty} (1 - \sigma_B) \sigma_B^i [\Lambda_{t,t+i+1} N_{f,t+i+1}], \quad (19)$$

where future cash flows are discounted by the household stochastic discount factor $\Lambda_t \equiv \beta(\lambda_t/\lambda_{t-1})$. I introduce a moral hazard friction similar to [Gertler and Kiyotaki \(2010\)](#); every period the bank can run away with a fraction of assets for personal gain. In this case the bank goes bankrupt and exits the industry. Fraud risk places a cap on the amount of deposits households are willing to credit to financial intermediaries. As a result, households rollover deposits only to the extent that banks find it sub-optimal to divert assets. The costly enforcement problem can be formally expressed as

$$V_{f,t} \geq \theta \left[q_t S_{f,t}^{H,b} + \Delta Q_t B_{f,t}^{L,H,b} + \eta RE_{f,t} \right]. \quad (20)$$

Equation (20) implies that in the margin, banks should be indifferent between operating honestly and diverting assets if the franchise value of the bank (left hand side) compensates for the real gains of diversion (right hand side). In that sense, the constraint makes banks incentive-compatible.

I consider an equilibrium in which the costly enforcement constraint always binds. Let $\theta \in [0, 1]$ be the absconding fraction of private assets; it follows that $\theta\Delta$ and $\theta\eta$ are the divertible shares of long term bonds and central bank reserves, respectively. Moreover, I assume that $\Delta, \eta \in [0, 1)$ to reflect advanced information with regards the share of government securities and central bank reserves in the bank balance sheet, which makes those assets harder to divert.

The value of the bank can be written recursively as

$$V_{f,t} \left(S_{f,t}^{H,b}, B_{f,t}^{L,H,b}, RE_{f,t}, D_{f,t} \right) = \max E_t \left\{ (1 - \sigma_B) \Lambda_{t,t+1} N_{f,t+1} + \sigma_B \Lambda_{t,t+1} V_{f,t+1} \left(S_{f,t+1}^{H,b}, B_{f,t+1}^{L,H,b}, RE_{f,t+1}, D_{f,t+1}^H \right) \right\}. \quad (21)$$

Each period, individual banks choose the sequence $\left\{ S_{f,t}^{H,b}, B_{f,t}^{L,H,b}, RE_{f,t}, D_{f,t} \right\}_{t=0}^{\infty}$ to maximize the franchise value subject to the balance sheet and the costly enforcement constraints.

As in [Gertler and Kiyotaki \(2010\)](#) I guess that the value function is linear

$$V_{f,t} \left(S_{f,t}^{H,b}, B_{f,t}^{L,H,b}, RE_{f,t}, D_{f,t} \right) = v_{S_t^{H,b}} S_{f,t}^{H,b} + v_{B_t^{L,H,b}} B_{f,t}^{L,H,b} + v_{RE_t} RE_{f,t} - v_{D_t} D_{f,t}. \quad (22)$$

Coefficients $v_{S_t^{H,b}}$, $v_{B_t^{L,H,b}}$, and v_{RE_t} are the marginal values of supplying an extra unit of credit to non-financial firms, holding an extra unit of long term bonds, and an extra unit of central bank reserves; v_{D_t} is the marginal cost of deposits.

The first order conditions are

$$E_t \Lambda_{t,t+1} \Omega_{t+1} (R_{t+1}^k - R_{t+1}^D) = \frac{\nu_t^{cec}}{1 + \nu_t^{cec}} \theta \quad (23)$$

$$E_t \Lambda_{t,t+1} \Omega_{t+1} (R_{t+1}^L - R_{t+1}^D) = \frac{\nu_t^{cec}}{1 + \nu_t^{cec}} \theta \Delta \quad (24)$$

$$E_t \Lambda_{t,t+1} \Omega_{t+1} (R_{t+1}^{RE} - R_{t+1}^D) = \frac{\nu_t^{cec}}{1 + \nu_t^{cec}} \theta \eta, \quad (25)$$

where

$$\Omega_t \equiv (1 - \sigma_B) + \sigma_B E_t \left\{ \Lambda_{t,t+1} \Omega_{t+1} \left[(R_{t+1}^k - R_{t+1}^D) \phi_t + R_{t+1}^D \right] \right\} \quad (26)$$

$$\phi_t \equiv \frac{E_t \Lambda_{t,t+1} \Omega_{t+1} R_{t+1}^D}{\theta - E_t \Lambda_{t,t+1} \Omega_{t+1} (R_{t+1}^k - R_{t+1}^D)}. \quad (27)$$

Equations (23), (24), and (25) define the excess returns on private and government assets, and central bank reserves, over the cost of deposits. $\nu_t^{cec} \geq 0$ is the Lagrange multiplier on the costly enforcement constraint. When the constraint is slack (i.e. $\nu_t^{cec} = 0$) excess returns vanish as households engage in frictionless arbitrage. If the constraint binds (i.e. $\nu_t^{cec} > 0$), limits to arbitrage drive positive excess returns on private and government assets over the deposit rate.

Equations (26) and (27) define the marginal value of bank's net worth and the maximum amount of bank leverage consistent with the incentive constraint. Due to the linearity of the value function, leverage is independent of bank-specific characteristics and implies that households rollover deposits the higher are excess returns, and the lower is the fraction of assets that can potentially be seized.

Equations (23) and (24) jointly determine an arbitrage condition between private and government assets

$$E_t \Lambda_{t,t+1} \Omega_{t+1} (R_{t+1}^L - R_{t+1}^D) = \Delta E_t \Lambda_{t,t+1} \Omega_{t+1} (R_{t+1}^k - R_{t+1}^D), \quad (28)$$

which implies that, in equilibrium, risk-adjusted weighted excess returns are equalized.

To complete the optimization problem, I focus on an equilibrium in which the incentive constraint always binds. As such, bank- f 's leverage can be expressed as the ratio of the adjusted measure of intermediated assets over own net worth^{12,13}

$$\phi_t = \frac{q_t S_{f,t}^{H,b} + \Delta Q_t B_{f,t}^{L,H,b} + \eta R E_{f,t}}{N_{f,t}}. \quad (29)$$

Equation (29) is the leverage ratio whose equilibrium condition is given by (27), and highlights the role of the agency friction in driving an endogenous leverage ratio. This is because bank's leverage depends on its own net worth, which varies endogenously over time. Moreover, focusing on an equilibrium with $\eta < \Delta < 1$, (29) delivers another powerful prediction; exchanging long term bonds with central bank reserves in periods of financial distress eases financing constraints and promotes credit to non-financial firms.

Aggregation. So long as banks face the same leverage constraint, I can sum across individual banks to get an expression for the aggregate leverage ratio

$$\phi_t = \frac{q_t S_t^{H,b} + \Delta Q_t B_t^{L,H,b} + \eta R E_t}{N_t}. \quad (30)$$

It follows that the aggregate bank balance sheet satisfies

$$q_t S_t^{H,b} + Q_t B_t^{L,H,b} + R E_t = N_t + D_t. \quad (31)$$

¹²The numerator is the maximum amount of an adjusted measure of bank's intermediated assets, a fraction θ of which the bank can divert without violating the incentive constraint. This is the amount of assets that enters the bank's balance sheet.

¹³I can drop the f -index since linearity of the value function implies that leverage is always constant.

The law of motion of aggregate net worth can be obtained by aggregating the retained earnings of surviving banks and the startup funds endowed to new banks, which is a share ξ of final period assets of exiting banks

$$N_t = \sigma_B \left[(R_t^k - R_t^D) \frac{q_{t-1} S_{t-1}^{H,b}}{N_{t-1}} + (R_t^L - R_t^D) \frac{Q_{t-1} B_{t-1}^{L,H,b}}{N_{t-1}} + (R_t^{re} - R_t^D) \frac{RE_{t-1}}{N_{t-1}} + R_t^D \right] N_{t-1} + \xi \left[q_{t-1} S_{t-1}^{H,b} + Q_{t-1} B_{t-1}^{L,H,b} + RE_{t-1} \right]. \quad (32)$$

Monetary Authority Different from [Eichenbaum et al. \(2021\)](#) and [Minesso et al. \(2022\)](#), I provide a distinctive framework that models the central bank as financing its activities through the issuance of central bank reserves. The operational framework is similar to that studied by [Sims and Wu \(2021\)](#) with a major difference: I add two new instruments for the central bank, the central bank digital currency and physical cash ([Chiu et al., 2023](#)).

In my model, relative to [Minesso et al. \(2022\)](#), the central bank implements conventional monetary policy, and intervenes in private and government asset markets during periods of financial distress. This structure makes it feasible to assess (i) the extend to which the introduction of a central bank digital currency influences the international transmission of domestic conventional and unconventional policies in response to country-specific shocks, (ii) potential trade-offs faced by the domestic and foreign monetary authorities, and (iii) international monetary policy cooperation.

In "normal times", conventional monetary policy is implemented through an endogenous feedback [Taylor \(1993\)](#) rule with inertia ([Clarida et al., 2000](#))

$$\frac{R_t^{nom}}{R^{nom}} = \left(\frac{R_{t-1}^{nom}}{R^{nom}} \right)^{\rho_m} \left[\left(\frac{\pi_t}{\pi} \right)^{\phi_\pi} \left(\frac{Y_t^{tot}}{Y^{tot}} \right)^{\phi_y} \right]^{1-\rho_m} e^{\sigma_m \epsilon_t^m}, \quad (33)$$

where R_t^{nom} is the short term nominal rate, $0 \leq \rho_m < 1$ the degree of interest rate inertia, $\phi_\pi > 1$ and $\phi_y \geq 0$ the feedback reactions to inflation and output gap deviations from target levels, and σ_m the standard deviation of the monetary policy shock. Variables not indexed by time hold in steady state. Nominal and real interest rates are connected

through a Fisher relation

$$R_t = \frac{R_{t-1}^{nom}}{\pi_t}.$$

In the model, I assume that deposits are short term (real) bonds remunerated at the real risk free rate. It follows that in equilibrium $R_t = R_t^D = (R_{t-1}^{nom}/\pi_t)$.

Unconventional policy tools. In addition to conventional interest rate policy, the central bank intervenes in the government and private capital markets to soften agency frictions and stimulate financial intermediation. This is because tighter balance sheet constraints lead to positive equilibrium excess returns that increase the cost of private borrowing and hinder economic activity.

Central bank intermediation involves the purchase of long term government and private securities at the respective prices Q_t and q_t . Direct purchases create a mismatch on the central bank balance sheet, which is settled with an equal amount of newly issued reserves, cash, and central bank digital currency. I assume that the issuance of central bank money (i.e. cash and CBDC) displaces central bank reserves one-for-one. Consequently, the central bank balance sheet can be expressed as

$$q_t S_t^{H,cb} + Q_t B_t^{L,H,cb} = RE_t + DC_t + M_t. \quad (34)$$

The budget constraint of the central bank satisfies

$$\begin{aligned} T_t^{cb} + R_t^{DC} DC_{t-1} + R_t^{re} RE_{t-1} + M_{t-1} + q_t S_t^{H,cb} + Q_t B_t^{L,H,cb} \\ \leq R_t^k q_{t-1} S_{t-1}^{H,cb} + R_t^L Q_{t-1} B_{t-1}^{L,H,cb} + RE_t + DC_t + M_t, \end{aligned} \quad (35)$$

where T_t^{cb} is a lump sum payment to the government from any operating surpluses ([Sims and Wu, 2021](#)), and DC_t the aggregate amount of CBDC issued by the domestic central bank.

Clearing in the private capital and public bond markets takes place when the aggregate supply of private and government debt is held either by domestic and foreign households, domestic financial intermediaries, or the domestic central bank. As such, aggregation in

the market for public debt — short and long term — requires

$$B = B_t^H + B_t^{*F} \quad (36)$$

$$B^{L,H} = B_t^{L,H,h} + B_t^{L,H,b} + B_t^{L,H,cb}, \quad (37)$$

where B and $B^{L,H}$ are constant (real) levels of government debt issuances. The private capital market clearing condition is given by

$$S_t = S_t^{H,b} + S_t^{H,cb}. \quad (38)$$

Exploiting equations (37) and (38), and the definition of the leverage ratio (30), I can derive an adjusted measure of aggregate leverage

$$q_t S_t = \phi_t N_t + q_t S_t^{CB} + \Delta Q_t \left(B_t^{L,H,CB} + B_t^{L,H,h} - B^L \right) - \eta R E_t. \quad (39)$$

Equation (39) delivers two key insights. First, when the enforcement constraint binds ($\nu^{cec} > 0$), central bank intermediation expands total demand for private assets, with positive spillovers on banks net worth (due to higher asset prices) and the overall supply of credit. Second, not all security purchases by the central bank have the same effect; because the agency problem is stronger for private assets (i.e. $\theta > \theta\Delta$, $\Delta < 1$), intervening in the private capital market tends to be more effective than equivalent interventions in the market for government bonds.

To ease exposition, I refer to central bank purchases of long term government debt as **quantitative easing**, and of private equity as **credit easing**. Quantitative and credit easing policies can respectively be formulated as

$$B_t^{L,H,cb} = \phi_{b^L,t} B^{L,H} \quad (40)$$

$$S_t^{H,cb} = \phi_{s,t} S_t, \quad (41)$$

where $\phi_{b^L,t}$ and $\phi_{s,t}$ denote the corresponding shares in total assets the central bank invests in during each program, and follow an AR(1) process in the levels.¹⁴

CBDC design. In the baseline scenario, the central bank supplies unremunerated

¹⁴In this regard, the model is suitable for the study of open economy effects of national unconventional monetary policies. However, I leave this for future research.

CBDC until all demand is satisfied. In that sense, variations in the quantity of newly issued digital currency are sufficient to clear the market, at the given interest rate. It follows that $\forall t$ ¹⁵

$$R_t^{DC} E_t \pi_{t+1} = 1. \quad (42)$$

Fiscal Authority The Treasury levies lump sum taxes, uses transfers from the central bank, and issues new short and long term debt — in fixed supply — to finance government spending G , and the cost of servicing any outstanding debt. Government spending is assumed to be exogenous and constant as a share total output.

The budget constraint is given by

$$G + R_t B + R_t^L Q_{t-1} B^L = B + Q_t B^L + T_t + T_t^{CB}. \quad (43)$$

Lump sum taxes (T_t) residually adjust to maintain fiscal balance.

Production The supply side block follows the conventional treatment employed in multi-country open economy New Keynesian models. There are four types of non-financial firms operating in the economy. A perfectly competitive final output wholesaler aggregates home and imported (i.e. foreign) final goods, which then distributes domestically. Perfectly competitive retailers produce bundles of final goods by aggregating domestically-produced and imported final goods. Intermediate firms hire capital and labor to produce individual varieties that are sold within the home country and exported to the foreign economy. Market power allows intermediate producers to set the price of their individual varieties subject to a [Calvo \(1983\)](#) friction. Capital goods are produced by a representative capital firm which uses final output and any undepreciated capital to invest in new capital subject to convex adjustment costs. To the interest of space, I formally present all standard components in [Appendix A.1](#).

¹⁵The CBDC rate (R_t^{DC}) is defined in real terms, and is adjusted by inflation since the central bank can only set nominal rates.

2.2 Calibration

The model is calibrated at a quarterly frequency for the U.S (home country) and the Euro-Area (E.A, foreign country).¹⁶ Parameters are divided between conventional, CBDC and cash related, and those specific to the banking sector. Conventional parameters follow the estimated model of [Eichenbaum et al. \(2021\)](#) and standard values in the literature. CBDC and cash related parameters are based on the structural framework of [Minesso et al. \(2022\)](#). Parameters at the core of the banking system are all calibrated internally to target empirical moments in U.S data. Table 1 provides a summary of selected values for the home economy. I refer the interested reader to Appendix A.2 for a comprehensive list.

Table 1. Parameterization

Parameter	Description	Value	Parameter	Description	Value
Households					
β	Discount factor	0.9950	χ	Utility weight of labor	0.9816
$\bar{B}^{L,H,h}/B^{L,H}$	Share of long term bonds	0.7500	φ^{-1}	Labor supply elasticity	1.0000
Financial intermediaries					
θ	Fraction of divertible assets	0.4126	ξ	Transfer to new bankers	0.0034
σ_B	Survival rate of bankers	0.9750	Δ	Seizure rate of long bonds	0.5000
Monetary Policy					
ϕ_π	Inflation response	1.6800	ϕ_y	Output response	0.0400
ρ_m	Interest rate smoothing	0.8700			
CBDC & Cash					
χ_{DC}	Weight of CBDC in utility	0.5000	ρ_{DC}	Substitution between payment instruments	2.0000
$\mu^\$$	Velocity of cash	0.5000	μ^{DC}	Velocity of CBDC	0.5000
$\xi^\$$	Storage cost of cash	1.0000	Γ	Median household preferences	0.5000

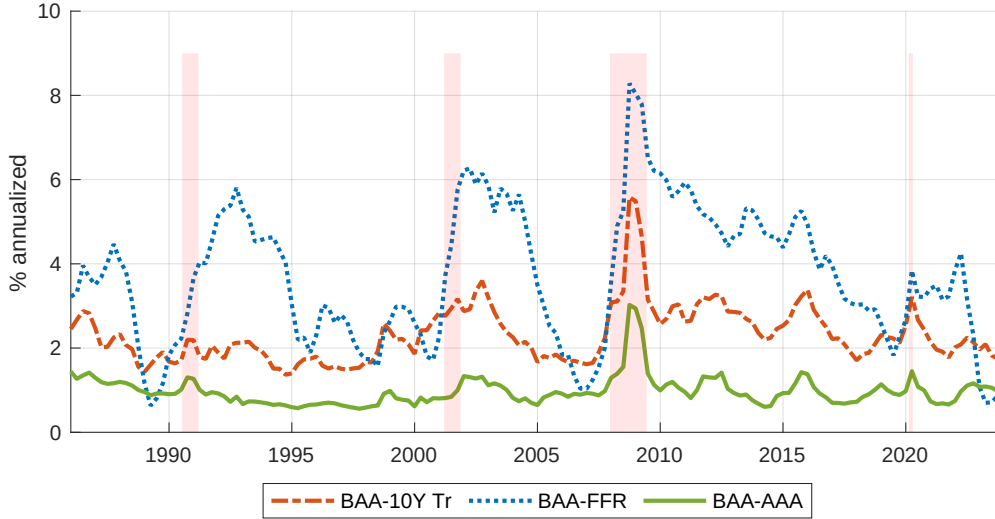
Starting off with the conventional parameters $(\beta, \chi, \varphi, \phi_\pi, \phi_y, \rho_m, \bar{B}^{L,H,h}/B^{L,H})$, the household discount factor is set to $\beta = 0.9950$ to allow for a 2% annualized net real interest rate in steady state. The utility weight of labor is set to $\chi = 0.9816$ to normalize steady state labor supply to one, and the inverse Frisch elasticity is set to $\phi = 1$. Taylor rule coefficients for inflation and output are set to $\phi_\pi = 1.68$ and $\phi_y = 0.04$, and the degree

¹⁶Germany can be thought of as the most representative country to approximate the E.A core.

of interest rate smoothing to $\rho_m = 0.87$. Based on the U.S long-run average distribution of long-maturity government debt, I assume that households hold 75% of the total supply of long term government bonds in steady state. It follows that $\bar{B}^{L,H,h}/B^{L,H} = 0.75$.

In spite of the expanding theoretical literature on the macroeconomic effects of introducing a central bank digital currency, empirical estimates of its structural attributes remain limited. One notable exception is [Li \(2023\)](#), who utilizes a structural demand model for Canada to estimate the share of CBDC in the total demand for liquid assets. In the absence of similar estimates for my benchmark economy, I adopt the calibration of the relevant parameters $(\mu^{DC}, \mu^{\$}, \chi_{DC}, \Gamma, \rho_{DC}, \xi^{\$})$ as outlined in [Minesso et al. \(2022\)](#). Specifically, the share of CBDC and cash in the liquidity aggregator is set to $\mu^{DC} = \mu^{\$} = 0.5$, displaying analogous functionality when used as payment instruments. Similarly, the weight of CBDC in the utility function is set to $\chi_{DC} = 0.5$, and the median household preference to $\Gamma = 0.5$.¹⁷ The elasticity of substitution between cash and CBDC is set to $\rho_{DC} = 2$. In the baseline simulation it is assumed that households can hoard cash costlessly. This implies that $\xi^{\$} = 1$.

Figure 1. Security spreads in the U.S (1986-2024)



NOTES. AAA and BAA are Moody's Seasoned AAA and BAA Corporate Bond Yields; 10Y Tr is the Market Yield on U.S Treasury Securities at 10-Year Constant Maturity; FFR is the Federal Funds Effective Rate. Light pink shaded lines are the U.S recession bands. Source: Federal Reserve Bank of St Louis.

The parameters internally calibrated are those specific to the banking system and the

¹⁷This is roughly equivalent to the upper bound of the estimate for CBDC demand in [Li \(2023\)](#).

principal agent friction underlying financial intermediation $(\sigma_B, \xi, \theta, \Delta)$.¹⁸ The survival probability of bankers is calibrated to $\sigma_B = 0.9750$ to reflect an average operational lifespan of ten years for each bank. The share of startup funds transferred to new bankers is calibrated to $\xi = 0.0034$ to match a steady state bank leverage ratio of 4, in line with much of the literature (Gertler and Karadi, 2013; Sims and Wu, 2021; de Groot and Haas, 2023). The fraction of total assets that banks can divert (θ), and the seizure rate of government bonds (Δ) are calibrated to reconcile with the following two targets: a credit spread of 100 basis points, and a bond spread of 50 basis points. In the model, the credit spread is defined as the difference between the return on effective capital and the short-term policy rate (Federal Funds Rate). In steady state, I approximate it using the empirical mean spread between the BAA and AAA corporate bond yields, drawn from the corresponding time series in figure 1. This choice is justified for the two alternative measures commonly used in the literature, namely the spread between the BAA and the 10 Year Treasury ("BAA - 10Y Tr"), and the spread between the BAA and the Federal Funds Rate ("BAA - FFR"), are likely to contain liquidity and term premia arising from a maturity mismatch, hence, not effectively isolating differences in the risk profile of the assets (de Groot and Haas, 2023).

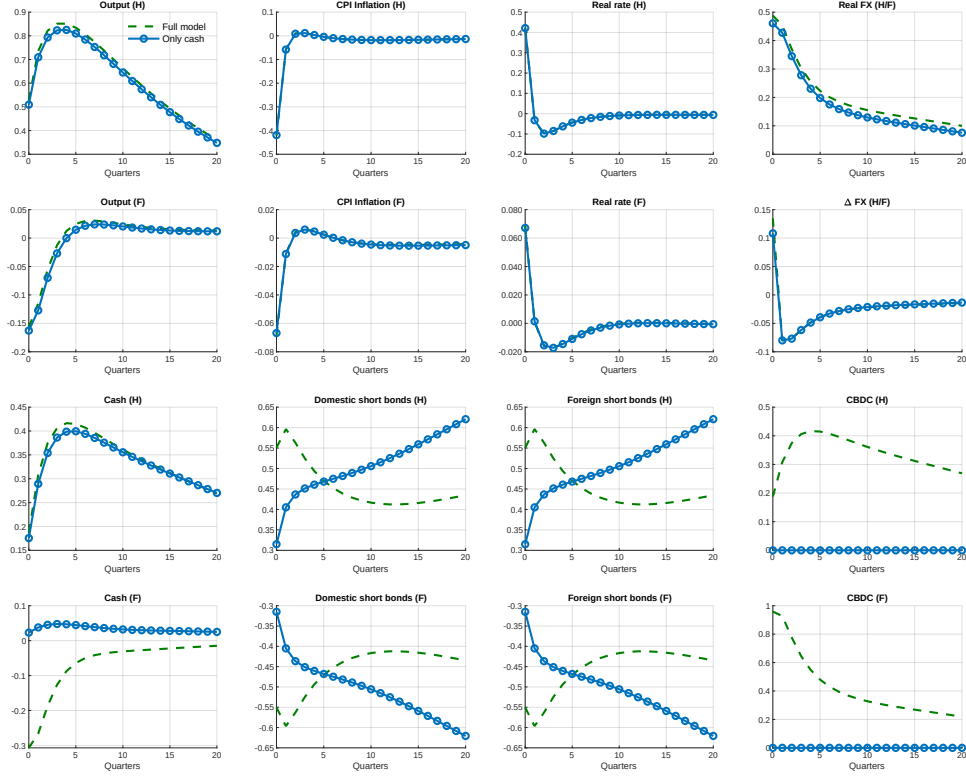
¹⁸As in Dedola et al. (2013), I assume that banks are symmetric across countries.

3 RESULTS

I demonstrate the dynamic propagation of country-specific shocks and the international transmission under financial frictions. To highlight the transmission mechanisms, I first consider the case where the economy operates only with cash and compare it against the counterfactual economy with CBDC.

3.1 Impulse responses

Figure 2. Transmission | Expansionary TFP



NOTES. Impulse responses of selected variables to a one standard deviation expansionary domestic total factor productivity shock. Values are in absolute deviations from the steady state.

The focus is on the (limiting) case of complete symmetry in the banking sectors across individual countries. This is feasible by equalizing the steady-state bank leverage and security spreads, which, in turn, implies that banks are equally capitalized and face the same agency friction.

Consider first the effects of a positive domestic TFP shock, in the absence of CBDC. Figure 2 presents the dynamic adjustment of the variables of interest. The rise in domestic total factor productivity increases the marginal productive capacity of private inputs (i.e. capital and labor), and subsequently total output in the domestic economy. This is followed by a reduction in the labor input as firms can accommodate higher demand with less labor.¹⁹ Lower employment puts a downward pressure on marginal costs, enabling firms to lower prices. Lower domestic prices (deflation) imply a depreciation of the domestic terms of trade (see A.3.1).²⁰ As inflationary pressures ease, the domestic central bank cuts the nominal rate. Due to policy inertia, the fall in inflation more than offsets the fall in the nominal rate, leading to a rise in the real interest rate. Savings become more attractive, inducing domestic households to increase their holdings of domestic bonds. The favorable economic outlook comes with positive effects on the demand for cash (due to the larger volume of transactions), domestic demand, and consumption and investment spending (see A.3.1). Holdings of foreign short term bonds also increase (see A.3.1). The domestic expansion generates spillover effects to the rest of the world. A deterioration in the terms of trade and the implied rise in the relative price of imports (from the domestic economy's perspective) influences international trade dynamics. While higher consumption and investment tend to fuel import demand, the terms of trade effect dominates, leading to an overall decline in foreign exports (i.e. domestic imports). It also contributes to deflation abroad — albeit less severe than in the home country — as imports (from the foreign economy's perspective) become cheaper. The relatively milder foreign deflation leads to a depreciation of the real exchange rate. The foreign economy experiences large capital inflows primarily driven by the higher demand for foreign bonds by households in the domestic economy. These inflows stimulate private investment and consumption in the foreign economy (see A.3.1). The latter is consistent with the observed increase in foreign cash holdings directed toward higher consumption spending. However, the decline in exports outweighs the positive effects of increased investment and consumption, ultimately causing a reduction in foreign total output. This, in turn, makes foreign households reduce their holdings of domestic and foreign bonds. Similarly, domestic demand in foreign economy falls due to higher prices on goods produced domestically (see A.3.1).

¹⁹The decline in hours worked in response to technological advancements aligns with the empirical findings in [Gali \(1999\)](#).

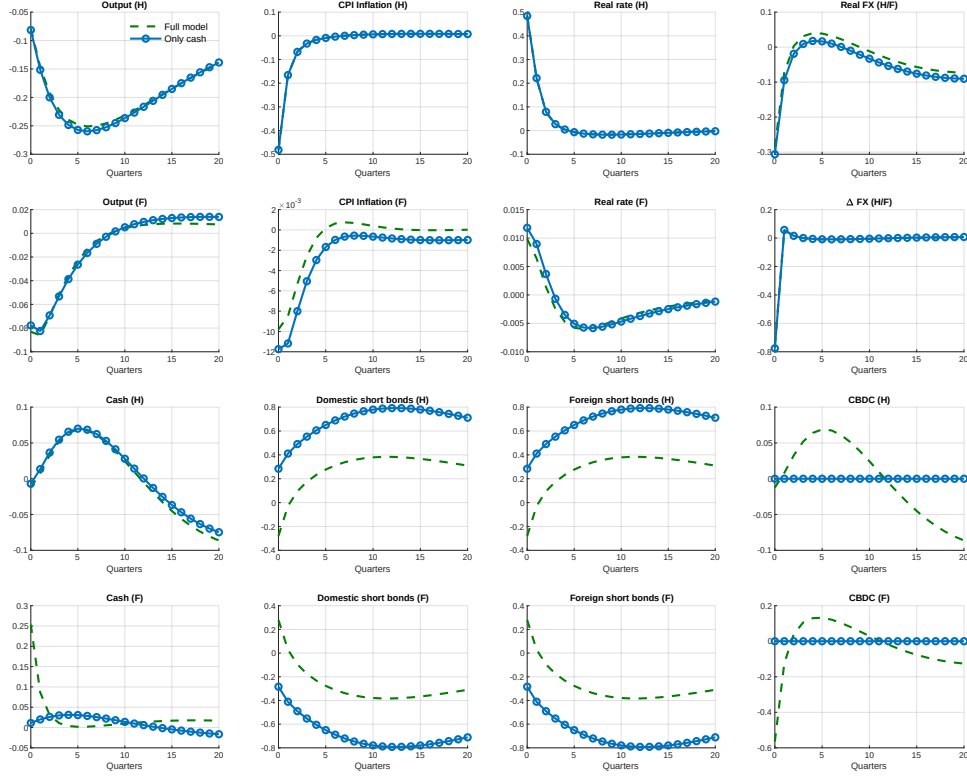
²⁰I define the domestic terms of trade as the ratio of import prices to export prices.

An important prediction of my model is that introducing CBDC *does* materially change the cross-border spillovers of the same total factor productivity shock. This confirms the core predictions of the "super charged" interest parity condition in [Minesso et al. \(2022\)](#), which points toward strong amplification effects. The same condition operates in my model,²¹ and is activated on impact, influencing the time paths of key macroeconomic aggregates. More specifically, nominal and real exchange rates exhibit non-trivial depreciations to the positive domestic TFP shock. Compared to the economy without CBDC the former increases by 24% and the latter by 6%, on impact. The stronger real exchange rate depreciation spillovers to foreign total output and internal demand, which rise by 3.8% and 9%, respectively. Moreover, despite the negative valuation effects from a nominal exchange rate appreciation (i.e. an appreciation of foreign currency, or, equivalently, a depreciation of the currency in which CBDC is remunerated) — somewhat paradoxically — foreign households increase their CBDC holdings. The intuition is that as domestic households become richer, they increase demand for foreign-currency denominated bonds, in addition to domestic short term debt. This tends to put a downward pressure on the foreign interest rate. Foreign households respond by running down on domestic short bonds due to lower interest payments, which tends to move the foreign real rate in the exact opposite direction. Overall, the two effects cancel each other out, and the foreign interest rate drops as foreign holdings of CBDC increase to pick up the slack from the shortfall in short term bonds. As such, so does the real exchange rate, effectively validating the amplification predicted by the super charged parity relation.

Consider now the effects of a contractionary conventional monetary policy shock in the domestic economy, in the absence of CBDC. Figure 3 presents the main impulse responses.

²¹From the foreign economy's perspective, one can get the super charged arbitrage equation by combining the first order conditions for domestic bonds and CBDC.

Figure 3. Transmission | Contractionary MP



NOTES. Impulse responses of selected variables to a one standard deviation contractionary monetary policy shock. Values are in absolute deviations from the steady state.

An increase in the central bank's policy rate triggers a domestic recession, marked by a sharp initial rise in the real interest rate and a decline in total output and inflation. The contraction in output stems from the negative impact of a higher real rate on domestic consumption and investment spending, ultimately weakening overall home demand (see [A.3.2](#)). The domestic recession generates significant international (trade) effects. Foreign output and inflation decline, while the foreign real interest rate rises — challenging the findings of [Minesso et al. \(2022\)](#), who argue that tighter domestic monetary policy should boost foreign production. This discrepancy arises from a combination of nominal and real exchange rate appreciations; the nominal exchange rate appreciates due to the relatively stronger increase in the domestic real interest rate, which, in turn, induces an appreciation of the real exchange rate despite foreign inflation falling less than domestic inflation. Because of the relatively steeper domestic deflation, the domestic terms of trade depreciate, making foreign exports more expensive and reducing import demand

(from the domestic economy’s perspective). As a result, despite the positive effects of rising foreign investment (see [A.3.2](#)), the contraction in foreign exports ultimately lowers foreign total output. In addition, and consistent with the intertemporal substitution channel, as domestic and foreign real rates rise, domestic households buy more home and foreign short term bonds, promoting international financial integration.

An equivalent contractionary monetary policy shock in the economy with CBDC generates more diverse effects. In particular, both the real and nominal exchange rates experience milder appreciations relative to the case without CBDC — by 7% and 2%, respectively. Over time, the expected depreciation of the exchange rate makes foreign households reduce their CBDC holdings for they find it less attractive to hold CBDC denominated in a currency undergoing devaluation. Thus, foreign households partly swap CBDC for domestic short term bonds. Moreover, there are larger capital flows from abroad into the domestic economy, leading to a larger accumulation of foreign short-term debt in the portfolios of foreign households relative to the case without CBDC. This confirms the gradual decline of both domestic and foreign real rates. Cash holdings of foreign households increase.

3.2 Financial sector effects

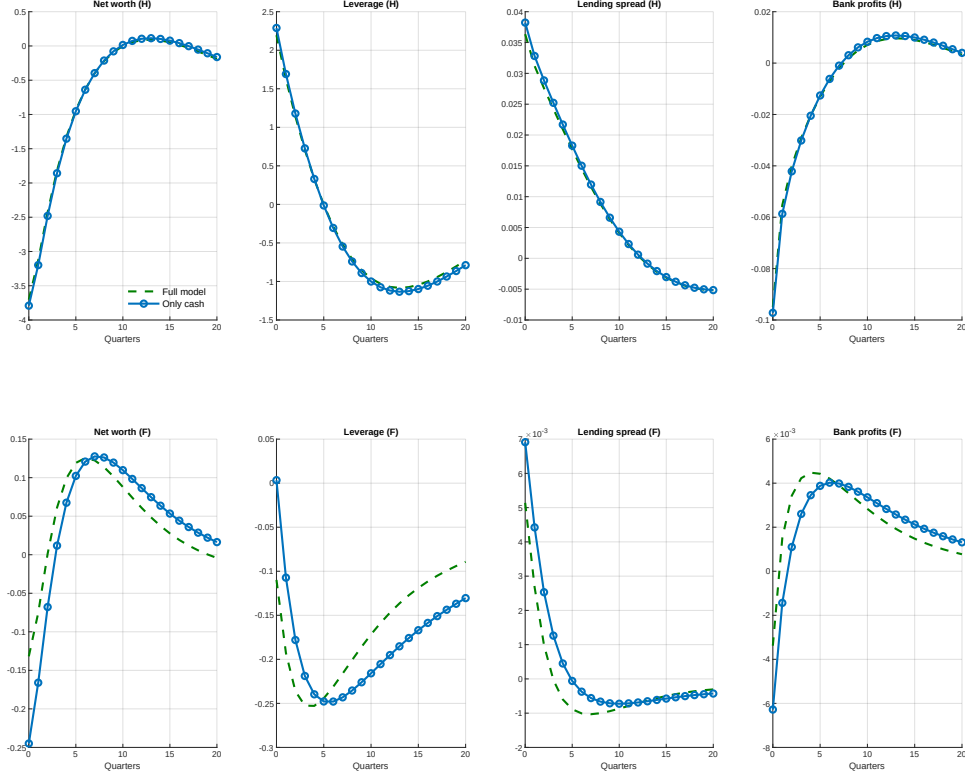
In this subsection, I consider how a productivity shock propagates to banks in the model without CBDC, and compare it against the model with a digital currency. The purpose of the experiment is to understand the extent to which issuing CBDC by one country may alter the international financial sector spillovers across banks, potential raising financial stability concerns. Impulse responses are summarized in [figure 4](#).

I begin by analyzing the transmission of a negative domestic TFP shock in the model without a CBDC, with a particular emphasis on the behavior of banking sector variables.²² A decline in domestic productivity weakens investment and capital accumulation, resulting in a fall in the price of capital and a lower realized return on capital. The deterioration in capital returns acts as a negative windfall shock for domestic banks, eroding their net worth and raising leverage ratios. In response, lending spreads widen, exacerbating the contraction in aggregate demand. This mechanism is consistent with

²²Since the model is solved linearly, a negative TFP shock generates impulse responses identical in magnitude but opposite in sign to a positive shock as considered before.

the fact that the initial downturn reduces both labor and asset income for households, which subsequently depresses bank deposits and further tightens financing conditions. It also contributes to lower bank profits.

Figure 4. Transmission | TFP & Financial Sector



NOTES. Impulse responses of selected banking variables to a one standard deviation negative total factor productivity shock. Variables are in absolute deviations from the steady state.

The effects are roughly similar for foreign banks. In particular, the strong co-movement in the demand for capital and investment induces a co-movement in the price of capital which depresses foreign bank net worth, raises bank leverage, and increases the foreign external financing premium. As such, the combination of financial frictions and the high degree of financial integration yields an international synchronization in the behavior of banking variables, in line with the findings in [Dedola and Lombardo \(2012\)](#).

At the core of my analysis is the beneficial role of CBDC for domestic and — most importantly — foreign banks. Specifically, the introduction of CBDC mitigates the decline in bank net worth and profit margins, while also dampening (or, even offsetting) the rise

in leverage and credit spreads. Notably, net worth of foreign banks falls by nearly 45% less compared to the scenario without a CBDC, while the increase in credit spreads and the decline in bank profits is approximately 28% and 50% smaller, respectively. Also, foreign banks become less levered (relative to the scenario without CBDC) by almost a factor of thirty. The idea is that in response to a negative domestic productivity shock, the model with CBDC produces a stronger real exchange rate appreciation, which increases the competitiveness of foreign producers, boosting exports towards the domestic economy, and ultimately foreign total output. This translates into higher investment in the foreign economy and a stronger appreciation of asset prices. The latter contributes to improving bank capitalization, effectively suppressing leverage and credit spreads, and increasing bank profitability.

4 CONCLUSION

In this paper I study the extend to which a new central bank policy tool, namely the introduction of a central bank digital currency, can affect the open economy landscape. To do so, I setup and solve a 2-country New Keynesian model with financial frictions and a realistic operational representation of monetary policy, in which the domestic central bank finances its operations through the issuance of a digital currency, in addition to cash and central bank reserves. In this setting, digital currency being available to both domestic and foreign households, has the power to shape the international transmission of domestic shocks. I first show that when accounting for a digital monetary instrument, the cross-border propagation of country-specific total factor productivity and conventional monetary policy shocks gets amplified relative to the counterfactual model without CBDC, thereby lending support to earlier studies that highlight the amplification effects of introducing a digital currency. I then show that in the model with CBDC, banks — both domestic and foreign — are more insulated against negative shocks to total factor productivity relative to the counterfactual economy without CBDC. The benefits are particularly pronounced for foreign banks due to the stronger appreciation of the real exchange rate that boosts foreign investment and appreciates asset prices to a greater extend than the model without a digital currency.

Arguably, an important limitation of this paper is that it does not explore the international spillovers of national unconventional monetary policy shocks — an issue that has remained at the forefront of policymakers’ agendas since the Great Financial Crisis. Given that the model incorporates financial frictions, such as limited commitment and segmented asset markets, which give real effects to central bank balance sheet interventions, it is well-suited for studying these type of unconventional monetary interventions. However, I leave this question for future research. Another limitation is the assumption of a constant debt path throughout the simulation period, effectively keeping fiscal policy inactive. This contrasts with the real-world practice, where fiscal policy plays a crucial role in cushioning economic shocks. Taken together, these limitations highlight the need for further research on monetary-fiscal policy interactions, including the study of both coordinated and uncoordinated conventional and unconventional policies.

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Appendix

A.1 Production sector

Final output wholesaler The final wholesaler chooses the quantity of the final good produced by the domestic final firm and consumed within the domestic economy Y_t^H , and the final good produced by the foreign final firm and exported to the domestic country X_t^{*F} (i.e. imported final good from the perspective of the domestic economy) to maximize profits, given by

$$\begin{aligned}\Pi_t^{AG} = & P_t^H \left[\int_0^\infty (Y_{j,t}^H)^{\frac{\nu-1}{\nu}} dj \right]^{\frac{\nu}{\nu-1}} - \int_0^\infty P_{j,t}^H Y_{j,t}^H dj \\ & + P_t^F \left[\int_0^\infty (X_{j,t}^{*,F})^{\frac{\nu-1}{\nu}} dj \right]^{\frac{\nu}{\nu-1}} - \int_0^\infty P_{j,t}^F X_{j,t}^{*,F} dj.\end{aligned}$$

$Y_{j,t}^H$ denotes the domestic intermediate variety j used by the domestic final firm in the production of Y_t^H ; $X_{j,t}^{*,F}$ is the foreign intermediate variety j used by foreign final firm as input in the production of X_t^{*F} ; ν is the elasticity of substitution between intermediate varieties; P_t^H stands for the price index of the final good produced and consumed domestically, and $P_{j,t}^H$ is the price of the domestic intermediate variety j ; consequently, P_t^F is the (export) price index of the foreign final good that is exported to the domestic economy, invoiced in units of domestic currency, and $P_{j,t}^F$ the corresponding price level of the foreign intermediate variety j (again, invoiced in units of domestic currency).²³

Optimization yields the following (optimal) demand functions for domestic and foreign intermediate varieties

$$Y_{i,t}^H = \left(\frac{P_{j,t}^H}{P_t^H} \right)^{-\nu} Y_t^H \quad (\text{A1})$$

$$X_{i,t}^{*F} = \left(\frac{P_{j,t}^F}{P_t^F} \right)^{-\nu} X_t^{*F}. \quad (\text{A2})$$

Substituting out in the original profit function, we get the price indexes

$$P_t^H = \left[\int_0^1 (P_{j,t}^H)^{1-\nu} dj \right]^{\frac{1}{1-\nu}}$$

²³In the model all exports are invoiced in units of currency of the destination country, i.e. the model only features local currency pricing.

$$P_t^F = \left[\int_0^1 (P_{j,t}^F)^{1-\nu} dj \right]^{\frac{1}{1-\nu}}.$$

Final bundlers Retailers combine domestic and imported goods into bundles of final private consumption, government consumption, and investment consumption goods, using a constant elasticity of substitution (CES) aggregator. Aggregate demands are given by

$$\begin{aligned} C_t &= \left[\omega^{1-\rho} (C_t^H)^\rho + (1-\omega)^{1-\rho} (C_t^F)^\rho \right]^{\frac{1}{\rho}} \\ G_t &= \left[\omega^{1-\rho} (G_t^H)^\rho + (1-\omega)^{1-\rho} (G_t^F)^\rho \right]^{\frac{1}{\rho}} \\ I_t &= \left[\omega^{1-\rho} (I_t^H)^\rho + (1-\omega)^{1-\rho} (I_t^F)^\rho \right]^{\frac{1}{\rho}}, \end{aligned}$$

where ω is the degree of home bias, and ρ the elasticity of substitution between domestic and foreign final goods.

The problem of each bundler in the domestic economy is to choose the amount of home and foreign final goods consumption to maximize profits. Optimization yields the following optimal demand functions

$$C_t^H = \omega \left(\frac{P_t^H}{P_t} \right)^{\frac{1}{\rho-1}} C_t \tag{A3}$$

$$C_t^F = (1-\omega) \left(\frac{P_t^F}{P_t} \right)^{\frac{1}{\rho-1}} C_t \tag{A4}$$

$$I_t^H = \omega \left(\frac{P_t^H}{P_t} \right)^{\frac{1}{\rho-1}} I_t \tag{A5}$$

$$I_t^F = (1-\omega) \left(\frac{P_t^F}{P_t} \right)^{\frac{1}{\rho-1}} I_t \tag{A6}$$

$$G_t^H = \omega \left(\frac{P_t^H}{P_t} \right)^{\frac{1}{\rho-1}} G_t \tag{A7}$$

$$G_t^F = (1-\omega) \left(\frac{P_t^F}{P_t} \right)^{\frac{1}{\rho-1}} G_t, \tag{A8}$$

where P_t is the consumer price index in the domestic economy, and is defined as

$$P_t = \left[\omega (P_t^H)^{\frac{\rho}{\rho-1}} + (1 - \omega) (P_t^F)^{\frac{\rho}{\rho-1}} \right]^{\frac{\rho-1}{\rho}}. \quad (\text{A9})$$

Let the domestic final good consumed within the domestic economy, and the foreign final good exported to the domestic economy be utilized for domestic private, government, and investment consumption

$$\begin{aligned} Y_t^H &= C_t^H + G_t^H + I_t^H \\ X_t^{*F} &= C_t^H + G_t^H + I_t^H. \end{aligned}$$

Substituting out the optimal demand functions, we get the total domestic demand for home-produced final goods

$$Y_t^H = \omega \left(\frac{P_t^H}{P_t} \right)^{\frac{1}{\rho-1}} (C_t + G_t + I_t), \quad (\text{A10})$$

and the total domestic (import) demand for foreign-produced final goods

$$X_t^{*F} = (1 - \omega) \left(\frac{P_t^H}{P_t} \right)^{\frac{1}{\rho-1}} (C_t + G_t + I_t). \quad (\text{A11})$$

Intermediate producers A continuum of monopolistic firms $j \in [0, 1]$ produce differentiated intermediate goods using capital $K_{j,t}$ and labor $L_{j,t}$. Technology is given by

$$Y_{j,t}^H + X_{j,t}^F = A_t K_{j,t}^\alpha L_{j,t}^{1-\alpha}, \quad (\text{A12})$$

where A_t denotes aggregate productivity, $X_{j,t}^F$ is domestic exports of the intermediate variety to the foreign economy, and α is the capital share. To rent capital, intermediate firms issue equity claims $S_{j,t}$ at price q_t , held directly by private banks and (potentially by) the central bank. The units of capital claims equal the number of units of acquired capital $K_{j,t+1}$. It follows that $q_t S_{j,t} = q_t K_{j,t+1}$.²⁴

²⁴Capital needs some time to build, thus becoming effective next period.

Imposing symmetry, cost minimization yields the demand for labor and capital

$$W_t = (1 - \alpha)MC_t A_t K_t^\alpha L_t^{-\alpha} \quad (\text{A13})$$

$$Z_t = \alpha MC_t A_t K_t^{\alpha-1} L_t^{1-\alpha}, \quad (\text{A14})$$

where MC_t is the real marginal cost which is independent of firm-specific characteristics, hence common across all firms.²⁵

Market power allows intermediate firms to set the price of their variety taking demand as given. I assume that firms face Calvo (1983) frictions when setting optimal prices to maximize the expected stream of (real) profits.

Optimization yields the optimal price for domestic sales ($\hat{P}_t^H$) and exports ($\hat{P}_t^{*F}$)

$$E_t \sum_{k=0}^{\infty} (\beta\zeta)^k \lambda_{t+k} \left[\frac{\hat{P}_t^H}{P_t} \frac{P_t}{P_{t+k}} \left(\frac{\hat{P}_t^H}{P_{t+k}^H} \right)^{-\nu} Y_{t+k}^H - \left(\frac{\nu}{\nu-1} \right) MC_{t+k} \left(\frac{\hat{P}_t^H}{P_{t+k}^H} \right)^{-\nu} Y_{t+k}^H \right] = 0 \quad (\text{A15})$$

$$E_t \sum_{k=0}^{\infty} (\beta\zeta)^k \lambda_{t+k} \left[\frac{NER_{t+k}}{NER_t} NER_t \frac{P_t^*}{P_t} \frac{P_t}{P_{t+k}} \frac{\hat{P}_t^{*F}}{P_t^*} \left(\frac{\hat{P}_t^{*F}}{P_{t+k}^{*F}} \right)^{-\nu^*} X_{t+k}^F - \left(\frac{\nu^*}{\nu^*-1} \right) MC_{t+k} \left(\frac{\hat{P}_t^{*F}}{P_{t+k}^{*F}} \right)^{-\nu^*} X_{t+k}^F \right] = 0. \quad (\text{A16})$$

The price index of home (intermediate) goods sold domestically (P_t^H) and exported (P_t^{*F}) can be defined as a weighted average of the newly reset (i.e. optimal) price and the price in effect in the previous period

$$P_t^H = \left[(1 - \zeta)(\hat{P}_t^H)^{1-\nu} + \zeta(P_{t-1}^H)^{1-\nu} \right]^{\frac{1}{1-\nu}} \quad (\text{A17})$$

$$P_t^{*F} = \left[(1 - \zeta)(\hat{P}_t^{*F})^{1-\nu^*} + \zeta(P_{t-1}^{*F})^{1-\nu^*} \right]^{\frac{1}{1-\nu^*}}. \quad (\text{A18})$$

Capital producers A representative capital producer buys the final good and any undepreciated capital, which uses to invest in the production of new capital. New capital is then sold to intermediate firms.

²⁵Constant returns to scale production function implies that the marginal cost and the capital-to-labor ratio do not depend on firm-specific characteristics, i.e. they are common across all firms. This, in turn, implies symmetric behavior across firms.

I assume that the evolution of investment is subject to adjustment costs ([Christiano et al., 2005](#)). The adjustment cost function takes the form

$$\Psi\left(\frac{I_t}{I_{t-1}}\right) = \frac{\phi^k}{2} \left(\frac{I_t}{I_{t-1}} - 1\right)^2,$$

where $\Psi(1) = 0$ and $\Psi'(1) = 0$, such that in steady state the adjustment cost has no effect. Additionally, the cost function is assumed to be convex, that is $\Psi''(1) > 0$. ϕ^k is the inverse elasticity of gross investment to the price of capital. The law of motion of the aggregate capital stock can be expressed as

$$K_{t+1} = (1 - \delta)K_t + I_t - \frac{\phi^k}{2} \left(\frac{I_t}{I_{t-1}} - 1\right)^2 I_t, \quad (\text{A19})$$

where δ is the capital depreciation rate.

Profit maximization yields the optimal demand for (gross) investment

$$q_t \left[1 - \frac{\phi^k}{2} \left(\frac{I_t}{I_{t-1}} - 1\right)^2 - \phi^k \left(\frac{I_t}{I_{t-1}} - 1\right) \frac{I_t}{I_{t-1}} \right] + E_t \Lambda_{t,t+1} \left[q_{t+1} \phi^k \left(\frac{I_{t+1}}{I_t}\right)^2 \left(\frac{I_{t+1}}{I_t} - 1\right) \right] = 1. \quad (\text{A20})$$

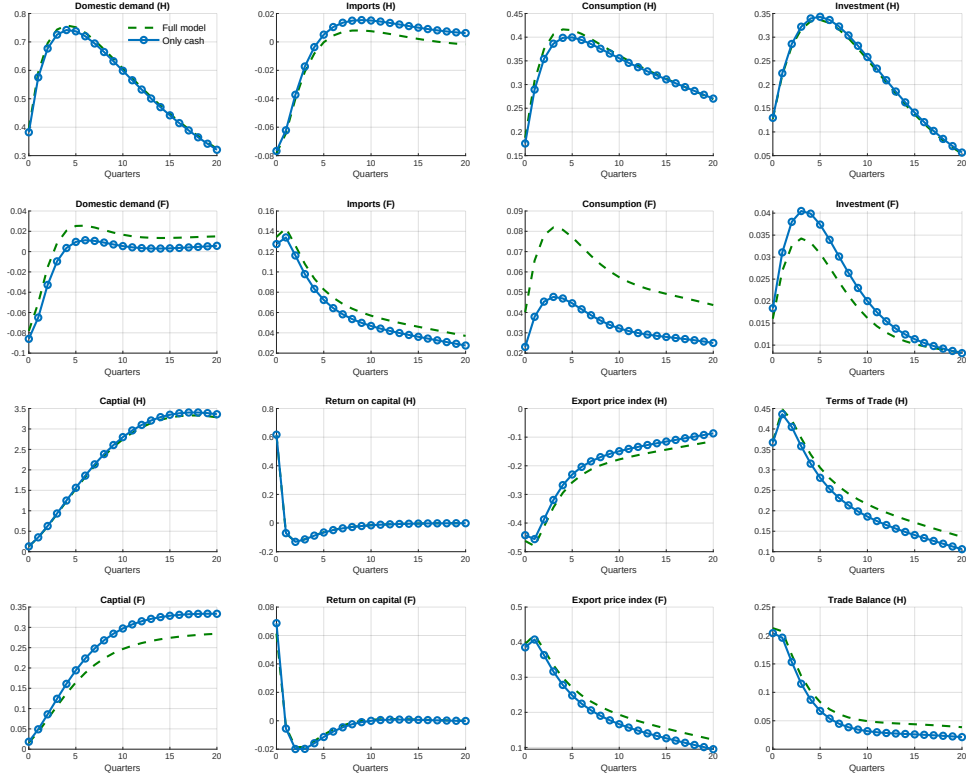
Note that any profits outside of the steady state are transferred lump sum to households.

A.2 Calibration

Domestic Economy			Foreign Economy		
Parameter	Description	Value	Parameter	Description	Value
β	Discount factor	0.9950	β^*	Discount factor	0.9950
h	Habit persistence	0.7500	h^*	Habit persistence	0.7500
φ^{-1}	Labor supply elasticity	1.0000	$(\varphi^*)^{-1}$	Labor supply elasticity	1.0000
ϕ^B	Cross-country portfolio cost	0.0010	ϕ^{B*}	Cross-country portfolio cost	0.0010
κ	Portfolio cost on long bonds	25.000	κ^*	Portfolio cost on long bonds	25.000
ξ^s	Storage cost of cash	1.0000	ξ^{s*}	Storage cost of cash	1.0000
ρ_{ec}	Persistence of pref. shock	0.8000	ρ_{ec}^*	Persistence of pref. shock	0.8100
σ_B	Survival rate of bankers	0.9750	σ_B^*	Survival rate of bankers	0.9750
η	Seizure rate of reserves	0.0000	η^*	Seizure rate of reserves	0.0000
Δ	Seizure rate of long bonds	0.5000	Δ^*	Seizure rate of long bonds	0.5000
ξ	Transfers to new bankers	0.0034	ξ^*	Transfers to new bankers	0.0027
θ	Absconding share of assets	0.4126	θ^*	Absconding share of asset	0.4126
α	Capital share	0.2500	α^*	Capital share	0.2500
ϕ^k	Investment adj. cost	5.2400	ϕ^{k*}	Investment adj. cost	3.8800
δ	Capital depreciation rate	0.0250	δ^*	Capital depreciation rate	0.0250
ζ	Calvo pricing	0.3300	ζ^*	Calvo pricing	0.7600
ν	Elasticity of interm. subs.	6.0000	ν^*	Elasticity of interm. subs.	6.0000
ρ	Elasticity of subs. between home and imported goods	1/3	ρ^*	Elasticity of subs. between home and imported goods	1/3
ρ_A	Persistence of TFP shocks	0.9300	ρ_A^*	Persistence of TFP shocks	0.9800
ϕ_π	Taylor rule inflation coeff.	1.6800	ϕ_π^*	Taylor rule inflation coeff.	1.4900
ϕ_y	Taylor rule output coeff.	0.0400	ϕ_y^*	Taylor rule output coeff.	0.2600
ρ_m	Interest rate smoothing	0.8700	ρ_m^*	Interest rate smoothing	0.9200
ρ_s	Persistence of CE shock	0.8000	ρ_s^*	Persistence of CE shock	0.8000
ρ_{bL}	Persistence of QE shock	0.8000	ρ_{bL}^*	Persistence of QE shock	0.8000
χ_{DC}	Weight of CBDC in utility	0.5000	χ_{DC}^*	Weight of CBDC in utility	0.5000
μ^s	Relative weight of cash	0.5000	μ^{s*}	Relative weight of cash	0.5000
μ^{DC}	Relative weight of CBDC	0.5000	μ^{DC*}	Relative weight of CBDC	0.5000
Γ	Median household pref.	0.5000	Γ^*	Median household pref.	0.5000
ρ_{DC}	Substitution between payment instruments	2.0000	ρ_{DC}^*	Substitution between payment instruments	2.0000

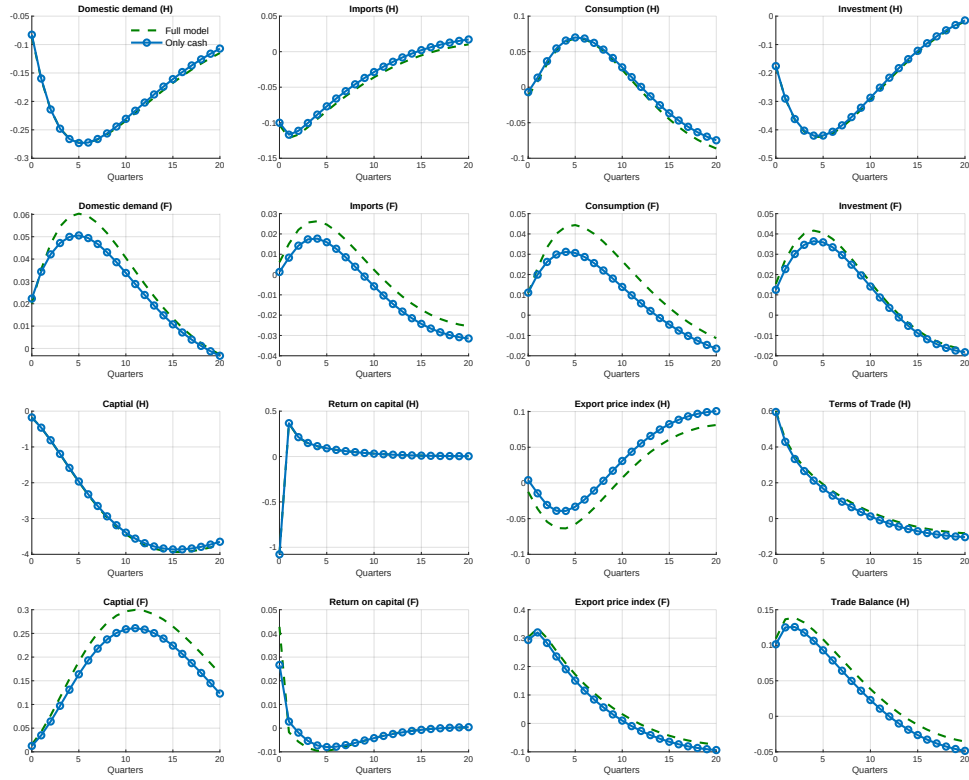
A.3 Additional results

Figure A.3.1. Transmission | Additional variables | Expansionary TFP



NOTES. Impulse responses of selected variables to a one standard deviation expansionary domestic total factor productivity shock. Variables are in absolute deviations from the steady state.

Figure A.3.2. Transmission | Additional variables | Contractionary MP

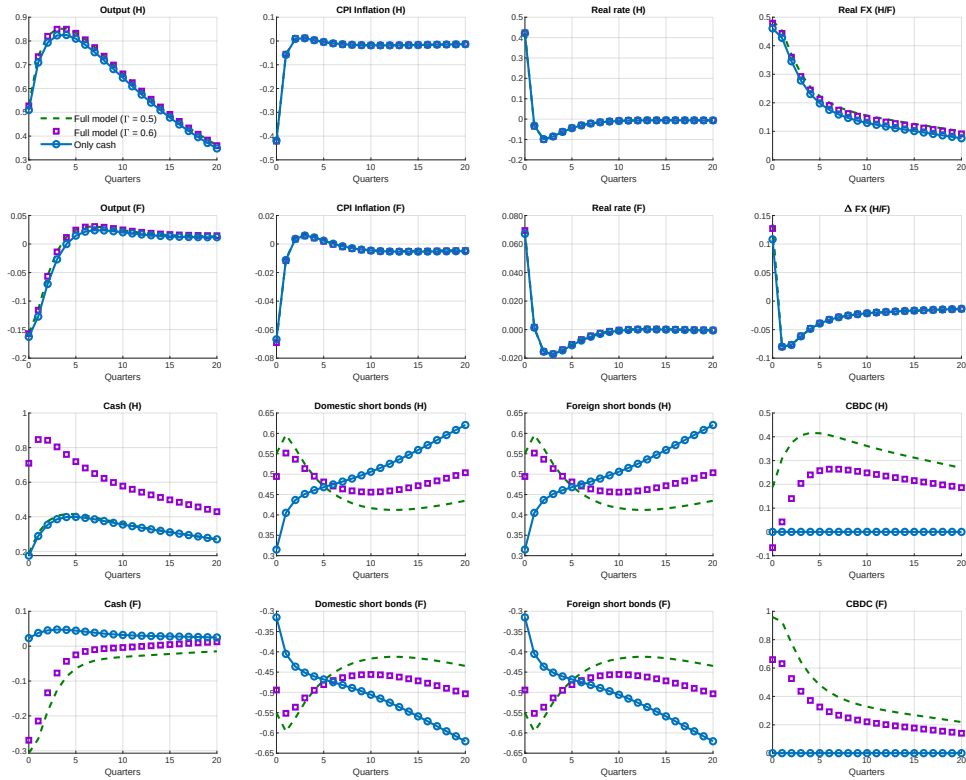


NOTES. Impulse responses of selected variables to a one standard deviation contractionary domestic monetary policy shock. Variables are in absolute deviations from the steady state.

A.4 Robustness

I test for the sensitivity of my findings with respect to variations in household preferences over cash and central bank digital currency. More precisely, I impose that households now prefer a less balanced bundle of liquid payment instruments, whereby they prioritize holding more cash and less digital currency. As such, I set $\Gamma = 0.600$, i.e. households prefer to have 60% of aggregate liquidity in the form of cash in hand.^{26,27}

Figure A.4.1. Robustness | Transmission | Expansionary TFP



NOTES. Impulse responses of selected variables to a one standard deviation expansionary domestic total factor productivity shock. Variables are in absolute deviations from the steady state.

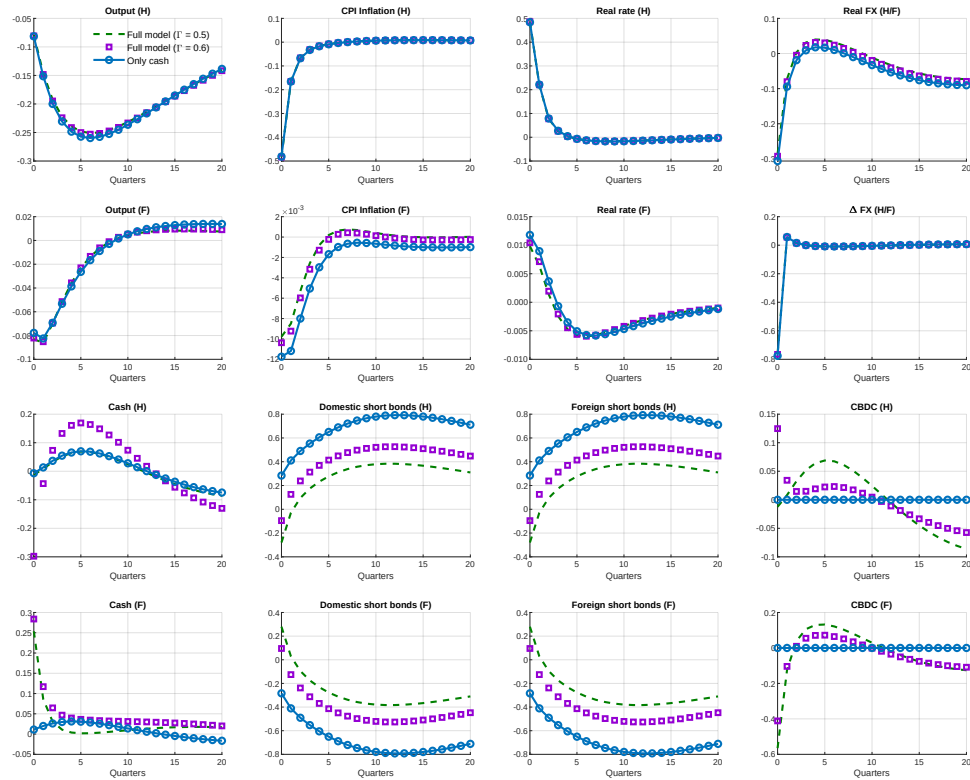
From figure A.4.1, one can see that for a domestic expansionary TFP shock, impulse response functions move in the same direction under both parameterizations. That is, they share the same qualitative properties, and it is only the magnitude of the initial response that differs. The same applies when the economy is subject to a contractionary

²⁶Recall that in the baseline model $\Gamma = 0.500$, i.e. households equally like cash and CBDC.

²⁷Due to the linearity of the model, assuming that $\Gamma = 0.400$ would simply rescale the magnitude of the impulse responses, without affecting the core predictions in the main text.

monetary policy shock, as one can see in figure A.4.2. As such, the main findings in the text remain robust to different splits between cash and central bank digital currency.

Figure A.4.2. Robustness | Transmission | Contractionary MP



NOTES. Impulse responses of selected variables to a one standard deviation contractionary domestic monetary policy shock. Variables are in absolute deviations from the steady state.